What are Operators?

Operators are **symbols** that perform operations on variables and values.
For example:

```
x + y   # '+' is an operator, and x, y are operands
```

⚙️ Types of Python Operators

Type	Description
1. Arithmetic Operators	Used for mathematical operations
2. Assignment Operators	Used to assign values
3. Comparison Operators	Compare two values
4. Logical Operators	Combine conditional statements
5. Identity Operators	Compare object identity
6. Membership Operators	Test for membership in sequences
7. Bitwise Operators	Operate on binary numbers

🔹 1. Arithmetic Operators

Operator	Description	Example
+	Addition	x + y
-	Subtraction	x - y
*	Multiplication	x * y
/	Division (float)	x / y
//	Floor Division	x // y
%	Modulus (remainder)	x % y
**	Exponent (power)	x ** y

```
x = 10
y = 3

print(x + y)   # 13
print(x - y)   # 7
print(x * y)   # 30
print(x / y)   # 3.333...
print(x // y)  # 3
print(x % y)   # 1
print(x ** y)  # 1000
```

- **2. Assignment Operators**

Operator	Example	Equivalent To
=	x = 5	x = 5
+=	x += 3	x = x + 3
-=	x -= 3	x = x - 3
*=	x *= 3	x = x * 3
/=	x /= 3	x = x / 3
//=	x //= 3	x = x // 3
%=	x %= 3	x = x % 3
**=	x **= 3	x = x ** 3

```
x = 5
x += 3    # 8
x *= 2    # 16
print(x)
```

🔹 3. Comparison Operators

Used to compare two values — result is **True or False**.

Operator	Description	Example	Result
==	Equal	x == y	False
!=	Not Equal	x != y	True
>	Greater than	x > y	True
<	Less than	x < y	False
>=	Greater or Equal	x >= y	True
<=	Less or Equal	x <= y	False

```
x = 10
y = 5
print(x == y)   # False
print(x != y)   # True
print(x > y)    # True
```

🔹 4. Logical Operators

Used to combine conditions.

Operator	Description	Example	Result
and	True if both are True	(x > 5 and y < 10)	True
or	True if at least one is True	(x > 5 or y > 10)	True
not	Reverses the result	not(x > 5)	False

```
x = 10
y = 5
```

```
print(x > 5 and y < 10)  # True
print(x > 5 or y > 10)   # True
print(not(x > 5))        # False
```

🔹 5. Identity Operators

Compare **object identity** (whether two variables refer to the same object in memory).

Operator	Description	Example
is	True if both variables point to same object	x is y
is not	True if not same object	x is not y

```
a = [1, 2, 3]
b = a
c = [1, 2, 3]

print(a is b)      # True (same object)
print(a is c)      # False (different objects, same value)
print(a is not c)  # True
```

🔹 6. Membership Operators

Check if a value exists in a sequence (like list, tuple, string, etc.)

Operator	Description	Example
in	True if value exists	"a" in "apple"
not in	True if value does not exist	"b" not in "apple"

```
fruits = ["apple", "banana", "cherry"]
print("banana" in fruits)     # True
```

```
print("grape" not in fruits)  # True
```

🔹 7. Bitwise Operators

Used for **binary (bit-level)** operations.

Operator	Description	Example	Result
&	AND	5 & 3	1
`	`	OR	`5
^	XOR	5 ^ 3	6
~	NOT	~5	-6
<<	Left shift	5 << 1	10
>>	Right shift	5 >> 1	2

```
x = 5   # 0101
y = 3   # 0011

print(x & y)   # 1
print(x | y)   # 7
print(x ^ y)   # 6
print(~x)     # -6
print(x << 1)  # 10
print(x >> 1)  # 2
```

📜 Summary Chart

Category	Example	Description
Arithmetic	+ - * / %	Math operations

Category	Example	Description
Assignment	= += -=	Assign values
Comparison	== != > <	Compare values
Logical	and or not	Combine conditions
Identity	is, is not	Object identity
Membership	in, not in	Check membership
Bitwise	`&	^ ~ << >>`